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ELA

1. Discuss various components of robot systems?

A robot system consists of.

Manipulator → Links & Joints

End effector → Gripper / tool

actuators → Motors

Sensors → Position, velocity, proximity

Controllers → Controls. robot motion

Processors → Excutes program

Software → Programming & control algorithms

2. Different types of end effectors used in robots?

Classification:

1) Mechanical Grippers. - Most common
   Two finger
   Three finger

2) Magnetic Gripper - Ferromagnetic parts

3) Vaccum Gripper - Suction cups for sheets

4) Adhesive Gripper - sticky surface handling

5) Hooks & scoops - Bulk materials

6) Active & passive Grippers

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3) Discuss Hydraulic & Pneumatic actuators in detail?

Hydraulic :-

Use pressurized oil

High power density

Used in Heavy robots

Components- pump, cylinder, valve, reservoir

Pneumatic:-

Use compressed air

Simple & Economical

Used in pick & place

low stiffness

Less accurate

4) Define forward & inverse kinematics of a robot?

Forward kinematics

Determines end-effector position from known joint angles

Input → Joint variables

output → position & orientation

Inverse kinematics

Determines Joint angles for a desired end-effector position.

Input → desired position

output → Joint angles.

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5) A frame F is moved by 1 unit along x-axis & 2 units along z-axis of reference frame find the new location of frame?

$$F = \begin{bmatrix} 2 & -1 & 1 & 3 \\ 1 & 2 & 3 & 2 \\ -1 & 0 & 2 & 4 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Given:-

From moved +1 along x

and +2 along z

Translation vector:-

$$T = \begin{bmatrix} 1 \\ 0 \\ 2 \\ 1 \end{bmatrix}$$

New position column = old position + Translation

$$= \begin{bmatrix} 3+1 \\ 2+0 \\ 4+2 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \\ 6 \\ 1 \end{bmatrix}$$

$$F' = \begin{bmatrix} 2 & -1 & 1 & 4 \\ 1 & 2 & 3 & 2 \\ -1 & 0 & 2 & 6 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

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